The Effects of Cumulative Workload on Player Physical Performance at the 2026 FIFA World Cup

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Introduction

As professional soccer players face increasingly crowded domestic, continental, and international match schedules, understanding how workload accumulation and lack of recovery time affect physical performance has become more critical. This issue is particularly relevant in international tournaments, where teams play several high-intensity matches within the span of a month. The 2026 FIFA World Cup provides a timely opportunity for studying this problem, as it is the first men's World Cup to feature an expanded 48 teams and 104 matches, reflecting the broader congestion of the professional soccer calendar. As fixture demands increase, understanding whether workload accumulated in earlier matches is associated with physical performance in subsequent matches may have practical implications for player rotation, recovery, and substitution decisions.

This project will investigate this relationship using player-level physical data from FIFA's 2026 World Cup post-match reports, which include measurements such as distance ran at various speeds, number of sprints, and top speed. Because players appear repeatedly over the course of the tournament, this data can be used to examine whether workload accumulated in earlier matches is associated with a player's physical output in subsequent matches. This project will focus on modeling these physical performance metrics using a mixed-effects model, which will enable the analysis to also account for systematic differences across players and teams while exploring the relationship between physical performance and prior workload and recovery.

Problem statement

Problem

The primary research question is: *How is physical performance impacted by prior match workload and recovery time among players at the 2026 FIFA World Cup?*

Methods

The primary statistical approach will be a linear mixed-effects model. The model will predict player physical performance in a match based on fixed effects such as prior workload, recovery time, and player position, as well as player and team-specific random intercepts to account for persistent differences in physical output across players and teams. Potential physical performance metrics include distance covered per 90 minutes, high-speed running distance per 90 minutes, and number of sprints per 90 minutes. Potential prior workload metrics include distance covered in the prior match, high-speed running distance in the prior match, and number of sprints in the prior match.

Data

The analysis will use player-level physical data from FIFA's 2026 World Cup post-match reports. The relevant variables (e.g. distance ran at various speeds, number of sprints, and top speed) will be extracted and organized into a player-match dataset, with one observation corresponding to a player's appearance in a particular match. The data will then be cleaned and used to construct longitudinal variables such as previous-match workload, cumulative tournament workload, recovery time, and normalized physical performance measures.

Deliverables

The main deliverables for the project will include:

- A cleaned dataset containing player-level physical performance measures from each match at the 2026 FIFA World Cup
- Exploratory analysis and visualizations of the player physical performance data
- A mixed-effects model that explores how player physical performance is impacted by prior workload and recovery time
- A final written report and project poster

Timeline

- Week 1: Status Update, Pre-Approval, Proposal
- Weeks 2-6: Extract player physical performance data from FIFA match reports, construct the player-match dataset, and derive longitudinal variables such as previous-match workload, cumulative tournament workload, and recovery time
- Weeks 7-9: Conduct exploratory data analysis and finalize the primary research question and model specification
- Week 10-11: Fit mixed-effects model and evaluate results
- Week 12: Write report and poster, submit draft to advisor
- Week 13: Revise the report and poster based on advisor feedback
- Reading week: Finalize report and submit